Thisuka Warsavithana

Full-Stack Developer | Backend & Systems

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Information Technology undergraduate specializing in full-stack web development and backend systems. Experienced in building and deploying end-to-end web applications with React, Next.js, Node.js, RESTful APIs and SQL/NoSQL databases, alongside Python and Gen AI work including RAG pipelines and LLM integrations. Seeking a full-stack software engineering internship.

Experience

Web & Technology Pillar Member — MoraSpirit Apr 2024 - Apr 2025

- Built responsive React UI components for sports.moraspirit.com and maintained the org's website and mobile app, including REST API integration.
- Collaborated across frontend and backend teams to keep the React interfaces consistent with MoraSpirit's branding.

Freelance Web Developer — Self-employed 2025 - Present

- Built hotel web profiles with Next.js, presenting rooms, facilities, photos and location details to attract guests.

Trainee Software Engineer — Institute of Computer Engineering Technology Sep 2023 - Feb 2024

- Developed the shopping cart module for a full-stack e-commerce application, handling add/update/remove operations and order totals.
- Built the inventory management backend, tracking stock levels and keeping product availability in sync with cart and order activity.
- Worked within an agile team, contributing through sprint planning, code reviews, and regular stand-ups.

Skills

Languages: TypeScript, JavaScript, Java, Python
Frontend: HTML5, CSS3, React, Next.js, React Native (Expo), Angular, Tailwind CSS, Bootstrap
Backend: Node.js, Express, RESTful APIs, Spring Boot, FastAPI, Socket.IO
Databases: PostgreSQL, MongoDB, IndexedDB (Dexie.js)
Testing: Node.js Test Runner (node:test), JUnit
AI/ML: AI API Integration (Google Gemini, Groq), scikit-learn, XGBoost, LangChain, FAISS, RAG, Sentence-Transformers
Tools & Platforms: Docker, Google OR-Tools, Prisma, Stripe, Firebase, Leaflet.js, Cloudinary, Git/GitHub, Postman, Jira

Education

B.Sc (Hons) in Information Technology — University of Moratuwa 2024 - 2028 (Expected)
CGPA: 3.40 / 4.00

Diploma in Software Engineering — Institute of Computer Engineering Technology 2023

G.C.E. Advanced Level — Physical Science — Ananda College, Colombo 10 2022 (sat 2023)
Z-score 1.89 · Combined Mathematics A · Chemistry A · Physics B · General English A

Projects

FreshRoute — Delivery Route Optimization Platform

React Native · Node.js/Express · TypeScript · Python/FastAPI · Google OR-Tools · PostgreSQL (Prisma) · Socket.IO · Stripe

- Built a VRPPD solver with Google OR-Tools, enforcing pickup-before-delivery and vehicle-capacity constraints, with a fallback heuristic for optimizer downtime.
- Built the driver mobile app (React Native/Expo) with live GPS tracking over JWT-authenticated Socket.IO connections and role-gated dispatch visibility.
- Deployed a Python microservice alongside the Node.js backend on Render; fixed a Prisma relation-mapping bug causing silent data loss.
- Led GitHub repo and hosting for a 5-person team — branch strategy, access control, and deployment.

github.com/ThisukaMW/freshroute-backend

ResolveX — Real-Time Delivery Fault Detection & Resolution Platform

FastAPI · React · Supabase (Postgres/Auth/Storage/Realtime) · scikit-learn · XGBoost · Google Gemini Vision

Champions — CodeArena'26 (organized by KDU and Dialog Axiata PLC)

- Trained and deployed four independent ML models (Random Forest classifier/regressor, Isolation Forest, calibrated XGBoost) to attribute delivery faults, predict ETA, and score refund-claim risk.
- Built a rule-governed Fairness Aggregator combining five evidence checks (timing, GPS route, zone, photo, claim history) into one of four explainable outcomes, with confidence-gated escalation to human review.
- Integrated the Google Gemini API for multimodal photo-evidence comparison and an in-app support assistant, with a deterministic rule-based fallback on API failure.
- Shipped real-time rider GPS tracking and automatic nearest-rider assignment across 6 role-based dashboards, deployed on Vercel.

github.com/ThisukaMW/ResolveX

RAG Document Q&A — Retrieval-Augmented Document Assistant

`Python · Streamlit · LangChain · FAISS · BM25 · HuggingFace (Sentence-Transformers) · Groq · MCP`

- Built a full RAG pipeline: page-aware PDF/TXT ingestion, chunking, local sentence-transformer embeddings, and a persistent FAISS index, served via a Streamlit chat UI with page-level citations.
- Implemented hybrid dense (FAISS) + sparse (BM25) retrieval via Reciprocal Rank Fusion, with a UI toggle to compare modes.
- Wrote a grounding-focused system prompt restricting answers to retrieved context, validated with adversarial test cases to prevent fabrication.
- Built an LLM-judged eval framework (4 question types) scoring 18/20 correctness and 20/20 faithfulness; exposed the knowledge base via an MCP server for external AI clients.
- Integrated the Groq LLM API (LangChain) with rate-limit handling and token/cost tracking; deployed on Streamlit Cloud.

github.com/ThisukaMW/rag-document-qa

GenZipher — Healthcare Security & Trust Platform

`Node.js · Express · MongoDB · React · Python`

Finalist — GenZipher 1.0 (organized by UCSC and CSSL)

- Built a healthcare platform preventing identity fraud and prescription and medicine tampering, with Ed25519-signed prescriptions behind a dual-signature dispense gate.
- Designed WebCrypto (ES256) device-binding auth and AES-256-GCM encryption for patient vitals, with break-glass access logging.
- Built a tamper-evident, hash-chained audit log and a Python anomaly-detection engine that scores risky users.
- Shipped 5 role-based dashboards with TOTP MFA and WebAuthn biometrics, deployed via Vercel/Firebase.

github.com/ThisukaMW/Genzipher

ResQ — Offline-First Disaster Response Platform

`React · Vite · Firebase (Auth/Firestore/Hosting) · Dexie.js (IndexedDB) · Leaflet.js`

- Designed a two-app system — offline-first field responder PWA and a real-time HQ dashboard — independently deployed on Firebase Hosting.
- Queued incident reports (GPS, photos, severity) to IndexedDB via Dexie.js, surviving airplane-mode and restarts with zero data loss.
- Built a multi-trigger foreground-sync system to work around iOS/Safari's unreliable background sync, with duplicate-upload prevention.
- Built a live Leaflet/OpenStreetMap incident map fed by Firestore onSnapshot listeners with duplicate-coordinate detection.

github.com/ThisukaMW/Disaster_Management

Achievements

- **CodeArena'26 Champions** — organized by KDU and Dialog Axiata PLC
- **TensorForge 2.0 Finalist** — AI Engineering & MLOps Buildathon, organized by KDU and Ascentic (Pvt) Ltd
- **GenZipher 1.0 Finalist** — organized by UCSC and CSSL
- **DEVthon Finalist** — organized by the Leo Club of the University of Moratuwa
- **Hackathon Finalist** — organized by APIIT
- **Sri Lanka University Games Champions** — Basketball, 2025
- **Freshers' Meet Champions** — Basketball, University of Moratuwa
- **Best Defensive Player** — Freshers' Meet Basketball, University of Moratuwa
- **Best Pillar Member of the Month** — MoraSpirit

Leadership & Activities

Secretary: Mora Basketball, University of Moratuwa

Member: Mora Basketball Team, University of Moratuwa

Junior Prefect: St. Sebastian's College, Moratuwa

Member: Basketball Team, St. Sebastian's College

Languages

Sinhala (native) · English (fluent)

References

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